

VPA A2 S2 Assessment Pack

Victoria Park Academy — A2 Level Computer Science

Semester 2 Assessment Pack

Academic Year: 2025 – 2026 | Semester: 2

Exam Board: Cambridge International A-Level Computer Science (9618)

Purpose: Mock examinations, project assessment criteria, mark schemes, grade boundaries, and progression tracking for the final semester before A-Level exams.

1. Mock Examination Overview

Mock	Week	Paper	Duration	Marks	Format
Mock 2	Week 3	Paper 1 (Theory)	1h 45m	75	Written exam, no notes, calculator permitted
Mock 3	Week 6	Paper 2 (Programming)	2h	75	Written exam, no notes, calculator permitted
Mock 4	Week 15	Paper 1 (Theory)	1h 45m	75	Written exam, no notes, calculator permitted
Mock 5	Week 16	Paper 2 (Programming)	2h	75	Written exam, no notes, calculator permitted

2. Mock Exam 2 — Paper 1 (Theory)

Specification

Duration: 1 hour 45 minutes

Marks: 75

Format: Structured questions, short answers, extended responses

Content: AS + A2 theory topics (Data Representation, Hardware, Software, Communication, Security, AI, Ethics, Boolean Logic, Advanced Theory)

Conditions: Silent room, no notes, non-programmable calculator, black pen

Mark Scheme Structure

Question Type	Marks	Marking Approach
1-mark recall	1	Correct answer only. No partial credit.
2 - 3 mark describe	2 - 3	One mark per valid point. Must be in context.
3 - 4 mark explain	3 - 4	One mark per reason + one mark for development/elaboration.
4 - 6 mark compare	4 - 6	Two marks per valid comparison point (similarity + difference). Structure rewarded.
6 - 8 mark analyse	6 - 8	Breakdown of components (2 - 3 marks), relationships (2 - 3 marks), conclusion (1 - 2 marks).
8 - 12 mark evaluate	8 - 12	Introduction (1), points with evidence (4 - 6), counter-argument (2 - 3), conclusion (1 - 2). Balanced argument essential.

Sample Question (Evaluate — 10 marks)

Question: Evaluate the use of biometric authentication compared to password authentication for securing a banking application.

Mark Scheme:

Introduction (1): Define biometric authentication and password authentication. State the context (banking security).

Point 1 + Evidence (2): Biometrics are more secure because they are unique to the individual and cannot be easily shared or stolen. Example: fingerprint or iris scan.

Point 2 + Evidence (2): Passwords are cheaper and easier to implement. No special hardware required. Users are familiar with them.

Point 3 + Evidence (2): Biometrics have privacy concerns. If biometric data is compromised, it cannot be changed (unlike a password). Storage of biometric data raises GDPR issues.

Counter-argument (2): Biometrics can be spoofed (e.g., fake fingerprints). Multi-factor authentication (password + biometric) offers the best balance.

Conclusion (1): For high-security banking, biometrics offer superior security but should be combined with passwords for defence in depth.

3. Mock Exam 3 — Paper 2 (Programming)

Specification

Duration: 2 hours

Marks: 75

Format: Structured questions requiring pseudocode, Python code, algorithm design, trace tables, and explanations

Content: Programming constructs, data types, subroutines, file handling, arrays, OOP, algorithms, computational thinking

Conditions: Silent room, no notes, non-programmable calculator, black pen, pencil for diagrams

Mark Scheme Structure

Question Type	Marks	Marking Approach
Trace table	3 – 5	One mark per correct row/column. Must show all variable changes.
Pseudocode / Code	4 – 12	One mark per correct construct. Efficiency and structure rewarded. Syntax errors penalised only if logic is unclear.
Algorithm design	6 – 10	Decomposition (2), correct logic (3 – 4), appropriate data structures (2), efficiency (1).
Explanation of code	2 – 4	One mark per valid point. Must reference specific lines or variables.
Comparison of algorithms	4 – 6	Time complexity (2), space complexity (2), suitability for context (2).

Sample Question (Algorithm Design — 10 marks)

Question: A program stores student records in a text file. Each record contains: StudentID (string), Name (string), Score (integer). Write a program that reads the file, stores the data in an appropriate data structure, and allows the user to search for a student by ID using binary search. Output the student's name and score if found, or a suitable message if not found.

Mark Scheme:

Data structure (2): Array of records (or list of dictionaries) declared correctly. Fields match requirements.

File handling (2): File opened correctly, read line by line, parsed into fields, closed properly.

Binary search algorithm (4): Data sorted before search (1), lower/upper bounds initialised (1), loop with midpoint calculation (1), correct comparison and bounds adjustment (1).

Output (1): Correct output for found and not-found cases.

Structure and efficiency (1): Modular design, meaningful variable names, comments.

4. Mock Exam 4 — Paper 1 (Final Theory Diagnostic)

Specification

Duration: 1 hour 45 minutes

Marks: 75

Format: Full A-Level Paper 1 equivalent

Purpose: Final diagnostic of theory readiness before the real exam

Content coverage: All AS and A2 theory topics, weighted to syllabus proportions

Content Weighting

Topic	Approximate Marks	% of Paper
Data Representation	10 – 12	14 – 16%
Hardware & Processor Fundamentals	10 – 12	14 – 16%
System Software	8 – 10	11 – 13%
Communication & Internet	8 – 10	11 – 13%
Security & Privacy	8 – 10	11 – 13%
Artificial Intelligence	8 – 10	11 – 13%
Ethics & Legal Issues	6 – 8	8 – 11%
Boolean Logic & Advanced Theory	6 – 8	8 – 11%

5. Mock Exam 5 — Paper 2 (Final Programming Diagnostic)

Specification

Duration: 2 hours

Marks: 75

Format: Full A-Level Paper 2 equivalent

Purpose: Final diagnostic of programming readiness before the real exam

Content coverage: All programming and algorithm topics, weighted to syllabus proportions

Content Weighting

Topic	Approximate Marks	% of Paper
Programming Constructs	15 – 18	20 – 24%
Subroutines & Modular Design	12 – 15	16 – 20%
File Handling	8 – 10	11 – 13%
Arrays & Data Structures	10 – 12	13 – 16%

Topic	Approximate Marks	% of Paper
Object-Oriented Programming	8 – 10	11 – 13%
Searching & Sorting Algorithms	8 – 10	11 – 13%
Computational Thinking & Problem Solving	10 – 12	13 – 16%

6. Practical Project Assessment Criteria

The Practical Programming Project is assessed across six sections, totalling 75 marks.

Section	Max Marks	Bronze Target	Silver Target	Gold Target
Analysis	12	6 – 7	8 – 9	10 – 12
Design	15	7 – 9	10 – 12	13 – 15
Development	20	10 – 12	13 – 16	17 – 20
Testing	12	6 – 7	8 – 9	10 – 12
Documentation	8	4 – 5	5 – 6	7 – 8
Evaluation	8	4 – 5	5 – 6	7 – 8
Total	75	37 – 45	49 – 58	64 – 75

Detailed Criteria

Section	Criteria for Full Marks
Analysis	Clear, well-researched problem statement. Comprehensive requirements (functional and non-functional). Appropriate, measurable success criteria. Evidence of stakeholder consultation.
Design	Logical decomposition into modules. Appropriate algorithms with clear pseudocode/flowcharts. Justified data structures. Class diagram (if OOP). Data dictionary.
Development	Fully working solution. Efficient, well-structured code. Appropriate use of programming constructs. Evidence of iterative development (Git history). Good code organisation and comments.
Testing	Comprehensive test plan. Normal, boundary, and erroneous data. Evidence (screenshots). All tests linked to requirements. Failed tests identified, fixed, and re-tested.

Section	Criteria for Full Marks
Documentation	Clear technical guide (system design, data dictionary, annotated code). Usable user guide (installation, usage, screenshots, troubleshooting). Professional presentation.
Evaluation	Critical review against all success criteria. Honest identification of strengths and limitations. Stakeholder feedback gathered and responded to. Maintenance plan with examples. Future enhancements suggested.

7. Grade Boundaries

Grade boundaries are set based on historical Cambridge data and adjusted for mock difficulty. These are indicative boundaries for VPA mock exams.

Grade	Paper 1 (75 marks)	Paper 2 (75 marks)	Project (75 marks)	Total (225 marks)
A*	65+	65+	65+	195+
A	58 – 64	58 – 64	58 – 64	174 – 194
B	51 – 57	51 – 57	51 – 57	153 – 173
C	44 – 50	44 – 50	44 – 50	132 – 152
D	37 – 43	37 – 43	37 – 43	111 – 131
E	30 – 36	30 – 36	30 – 36	90 – 110
U	0 – 29	0 – 29	0 – 29	0 – 89

Note: Cambridge adjusts grade boundaries annually based on cohort performance. These boundaries are for internal tracking only and may be adjusted post-mock analysis.

8. Progression Tracking Template

Use this template to track each student's performance across all assessments in Semester 2.

Student	Target	Mock 2 (P1)	Mock 3 (P2)	Project	Mock 4 (P1)	Mock 5 (P2)	Current Total	Gap to Target	Intervention
Student A	A	62/75	58/75	64/75	65/75	60/75	309/375	On track	None
Student B	B	48/75	52/75	50/75	55/75	54/75	259/375	+6 marks	None

Student	Target	Mock 2 (P1)	Mock 3 (P2)	Project	Mock 4 (P1)	Mock 5 (P2)	Current Total	Gap to Target	Intervention
Student C	A	55/75	50/75	58/75	56/75	52/75	271/375	-17 marks	Paper 2 programming clinic
Student D	C	38/75	42/75	40/75	45/75	44/75	209/375	+8 marks	None
Student E	B	44/75	46/75	48/75	50/75	49/75	237/375	-6 marks	Theory revision pack

Intervention Codes

Code	Intervention	Target
INT-01	One-to-one theory clinic	Students scoring below target in Paper 1
INT-02	Programming practice pack	Students scoring below target in Paper 2
INT-03	Extended answer workshop	Students losing marks on 6+ mark questions
INT-04	Exam technique session	Students with good knowledge but poor exam performance
INT-05	Project support clinic	Students behind on project milestones
INT-06	Peer mentoring	Pair high-performing students with those who need support
INT-07	Parent contact	Students significantly below target or at risk of underachievement
INT-08	Wellbeing check	Students showing signs of stress or anxiety

9. Assessment Calendar

Week	Assessment	Action
Week 3	Mock 2 — Paper 1	Sit exam → Self-mark → Teacher review → Targeted intervention
Week 6	Mock 3 — Paper 2	Sit exam → Self-mark → Teacher review → Targeted intervention

Week	Assessment	Action
Week 9	Project Submission	Submit portfolio → Teacher marking → Feedback → Resubmission (if applicable)
Week 15	Mock 4 — Paper 1	Sit exam → Teacher marking → Individual feedback → Gap analysis
Week 16	Mock 5 — Paper 2	Sit exam → Teacher marking → Individual feedback → Final action plan
Week 17 – 18	Final Preparation	Revision clinics → Confidence building → Exam logistics briefing

10. Post-Mock Analysis Protocol

After every mock exam, follow this protocol to maximise learning:

Step 1 — Self-Assessment (Day 1): Students mark their own paper using the mark scheme. Complete a self-reflection sheet.

Step 2 — Teacher Marking (Day 1–2): Teacher marks all papers using the official mark scheme. Records marks per question in the tracking spreadsheet.

Step 3 — Class Analysis (Day 3): Teacher presents class-level data: average per question, common errors, high-scoring approaches.

Step 4 — Individual Feedback (Day 3–4): Each student receives a personalised feedback sheet with marks, comments, and a target action plan.

Step 5 — Targeted Intervention (Day 4–5): Students attend clinics, complete practice packs, or receive peer mentoring based on their gaps.

Step 6 — Retest (Following Week): Students complete a short retest on their weakest area to demonstrate improvement.

11. Resources

Cambridge A-Level Computer Science (9618) Syllabus (2024 – 2026)

Cambridge Past Papers (2018 – 2025) — Papers 1, 2, 3, and 4

Cambridge Mark Schemes (2018 – 2025)

Cambridge Examiner Reports (2018 – 2025)

VPA Bilingual Glossary (150 Terms) — Supporting Documents

VPA A2 Semester 2 Course Overview — Curriculum map and lesson structure

Document Control: Version 1.0 | Created: January 2026 | Owner: VPA Computer Science Department

Next Review: June 2026 (post-exam analysis)